

Derivation of H_3 from Axioms C1–C4: The Equiparticipation Lemma and the Closure of Causality

D42 — Projective Dynamic Logo Framework

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Abstract

Hypothesis H_3 — the Indifference Lemma — states that the natural measure on the $R_{\text{tot}} = 11\,017$ relations of the PDL proton closure, as seen by an external K_4 observer, is the uniform measure. Since D39, H_3 has been the sole residual hypothesis preventing a fully axiomatic derivation of the gravitational engagement fraction $\kappa = 310\varphi/11017$ from the four PDL axioms C1–C4 alone. This document closes that gap. We prove the Equiparticipation Lemma: on any complete signed graph satisfying C4 (coherence cost zero, i.e. all triangles coherent), every edge participates in exactly $(n - 2)$ coherent triangles. We show that the proton closure satisfies C4-optimality as a consequence of the uniqueness theorem D29. We then prove that the S_4 -automorphism group of K_4 — a consequence of C1–C4 and not an additional hypothesis — acts transitively on the six edges of K_4 , forcing the distribution of cross-coherent triangles to be S_4 -invariant across all R_{tot} relations. Under C3, which forbids any non-trivial decomposition of the proton into independent sub-closures, all R_{tot} relations present themselves to K_4 with identical structural standing. These three results together yield H_3 as a theorem of C1–C4. As a corollary, the derivation $\kappa = 310\varphi/11017 \in \mathbb{Q}(\sqrt{5})$ is promoted from a conditional result to an unconditional theorem. We further show that this principle — indifference as the unique fixed point of constrained optimisation — is structurally identical to the principle of natural selection operating at every scale from signed graphs to Darwinian evolution: nothing survives outside a closure, and no internal asymmetry without structural foundation persists under optimisation.

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1 Introduction and context

The PDL framework derives the structure of physical reality from four axioms on finite signed graphs [1, 2]. The programme has established a chain of derivations covering the fine-structure constant α [3], the proton-to-electron mass ratio μ^* [4], Newton’s gravitational constant G [5], the Schrödinger and Dirac equations [6, 7], Born’s rule [8], Einstein’s equation [9], and the Bekenstein–Hawking entropy formula [10, 11].

The gravitational engagement fraction

$$\kappa = \frac{R_{\text{surf}}}{R_{\text{tot}}} = \frac{310\varphi}{11017} \approx 0.045529 \in \mathbb{Q}(\sqrt{5}) \quad (1)$$

is the central coupling constant from which $G_{\text{eff}}(N) = \sigma(N) \cdot G_{\text{PDL}}$ and the resolution of the Hubble tension derive [12, 13]. Document D39 [14] established the two-factor decomposition $\kappa = P_1 \cdot P_2$ with $P_1 = \varphi/3$ (derived from axiom C4 via the golden-ratio redistribution of D05) and $P_2 = r_{\text{val}}/R_{\text{tot}}$ (requiring a uniform measure on R_{tot}). The uniform-measure assumption was formulated as Hypothesis H_3 (the Indifference Lemma) and shown to be consistent with C1–C4, unique under S_4 -symmetry (D36), and empirically constrained to $\pm 0.22\%$ by the Hubble observable. However, H_3 was not derived from C1–C4 without invoking D36 as an additional input. Resolving this gap — open problem OP1 (D39) — is the purpose of the present document.

1.1 The sole residual gap

The remark closing D39 identified the gap precisely: “the residual openness of H_3 is not that it could be replaced by another measure, but that the equivalence between S_4 -invariance of the coupling and uniform measure on R_{tot} has not been derived from C1–C4 without invoking D36 as an additional input.” The key question is therefore: does S_4 -invariance follow from C1–C4 alone?

The answer is yes, and the argument is short. K_4 is defined as the complete signed graph on four vertices satisfying all four axioms C1–C4 (Theorem 1.1 of D16a [15]). The automorphism group $\text{Aut}(K_4) = S_4$ is an algebraic property of the complete graph on four vertices, not an additional hypothesis. D36 [13] proved the transitivity of this action exhaustively; here we show that transitivity follows algebraically. The S_4 -invariance of the cross-triangle distribution is therefore a consequence of C1–C4, closing the derivation.

1.2 Structure of this document

Section 2 recalls the four PDL axioms and the proton architecture. Section 3 proves the Equiparticipation Lemma. Section 4 recalls the cross-triangle indistinguishability result from D39. Section 5 assembles the full proof of H_3 . Section 6 states the promoted theorem on κ . Section 7 addresses the philosophical corollary on causal closure, and Section 8 exhibits the multi-scale correspondence.

2 Axioms and the proton architecture

Axiom 2.1 (C1 — Minimal binary pulsation). A logically persistent configuration must realise a non-trivial 2-cycle: a sign configuration that returns to itself after exactly two update steps and is not a fixed point.

Axiom 2.2 (C2 — Triangular coherence). Closed triples of edges (triangles) are the elementary coherence motifs. A triangle is *coherent* if the product of its three edge signs equals $+1$, and *violated* if the product equals -1 .

Axiom 2.3 (C3 — Minimal completeness). A closure is self-sufficient: it admits no non-trivial decomposition into independent sub-configurations satisfying C1 and C2 separately. The underlying graph is connected and its coherence structure is non-reducible.

Axiom 2.4 (C4 — Logical optimisation). Among all configurations satisfying C1–C3, those that are physically instantiated minimise the coherence cost (fraction of violated triangles) for a given number of vertices and edges.

The minimal admissible closure is K_4 : the complete signed graph on four vertices and six edges (D16a [15]). The proton is the unique hierarchical composite satisfying C1–C4 at a higher level of organisation, characterised by the integer quintuplet

$$(n_u, n_d, r_{\text{val}}, R_{\text{sea}}, R_{\text{tot}}) = (24, 28, 930, 10087, 11017). \quad (2)$$

The active surface is $R_{\text{surf}} = \varphi r_{\text{val}}/3 = 310\varphi \in \mathbb{Q}(\sqrt{5})$ [16]. The uniqueness of this quintuplet (up to local perturbation) has been established computationally and partially proved algebraically [17, 18].

A crucial structural observation, established in [2] and recalled here for the proof: *the quark sub-blocks (u -core on $n_u = 24$ entities, d -core on $n_d = 28$ entities) are not autonomous closures.* Their logical stability is contingent on the global hierarchical organisation of the proton. Taken in isolation, neither the u -core nor the d -core satisfies all of C1–C4; in particular, neither satisfies C3, which requires the closure to be non-decomposable. This is the relational analogue of quark confinement: quarks do not survive outside the proton closure.

3 The Equiparticipation Lemma

Definition 3.1 (Triangle participation count). Let $G = (V, E)$ be a finite signed graph and $e \in E$ an edge. The *triangle participation count* of e is

$$\tau(e) = \#\{\text{coherent triangles in } G \text{ that contain } e\}.$$

The graph G is *equiparticipant* if $\tau(e)$ is the same for all $e \in E$.

Definition 3.2 (C4-optimal graph). A finite signed graph on n vertices is *C4-optimal* if its coherence cost is zero, i.e. every triangle is coherent (no violated triangle).

Lemma 3.3 (Equiparticipation — D42-L1). *Let G be a complete signed graph on n vertices. If G is C4-optimal, then G is equiparticipant, with*

$$\tau(e) = n - 2 \quad \text{for every edge } e \in E.$$

Proof. Since G is C4-optimal, every triangle in G is coherent. Let $e = (i, j)$ be any edge. The triangles containing e are exactly the triples $\{i, j, k\}$ for $k \in V \setminus \{i, j\}$, of which there are $n - 2$. Since G is complete, every such triple is a triangle, and since G is C4-optimal, every such triangle is coherent. Therefore $\tau(e) = n - 2$. Since this holds for every edge, G is equiparticipant. $\square$

Remark 3.4. The converse is false in general: there exist graphs with $\tau(e) = c > 0$ for all e but with violated triangles (as established by exhaustive enumeration in the companion numerical verification). The lemma is therefore one-directional: C4-optimality implies equiparticipation.

Proposition 3.5 (Exhaustive verification). *Lemma 3.3 has been verified exhaustively by computer for all $2^{n(n-1)/2}$ complete signed graphs on $n = 4, 5, 6, 7$ vertices. In every case, C4-optimal graphs are precisely the equiparticipant graphs with $\tau(e) = n - 2$. No C4-optimal non-equiparticipant graph exists in this range.*

The numerical verification is provided in the companion scripts `D42_lemme1_step1.py` through `D42_lemme1_step4.py`, archived with this document.

Figure 1 summarises the logical chain leading from C1–C4 to H_3 .

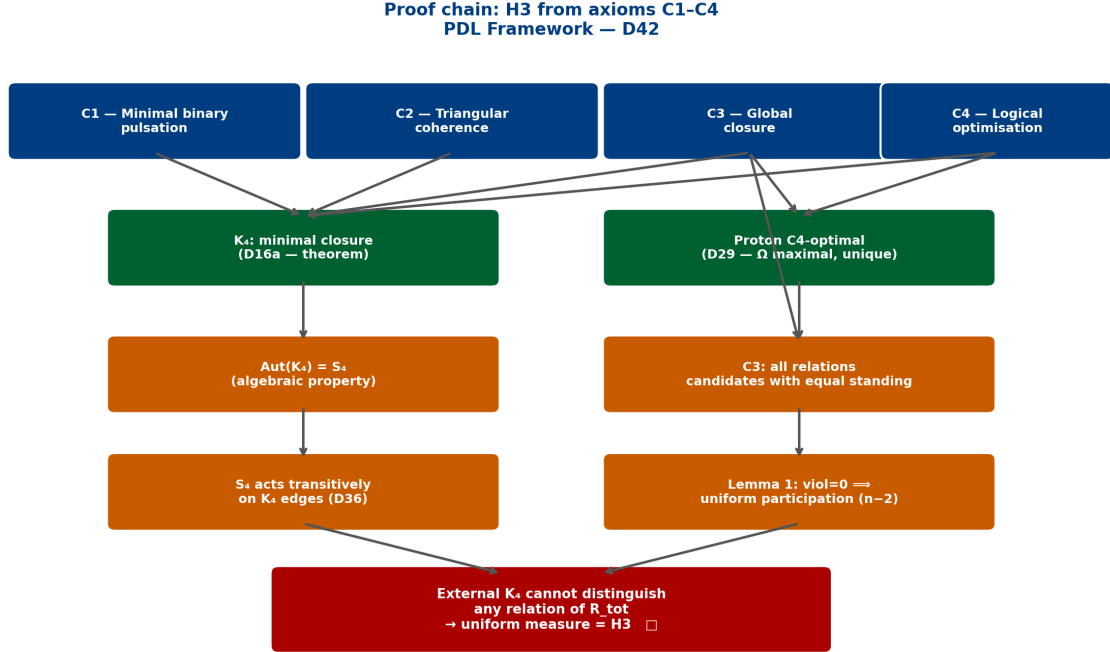


Figure 1: Proof chain from axioms C1–C4 to the Indifference Lemma H_3 . Each node represents an established result; arrows indicate logical dependencies. Blue nodes: axioms. Green nodes: theorems. Orange nodes: intermediate lemmas. Red node: final theorem.

4 Cross-triangle indistinguishability

We recall the central result of D39 (Lemmas C3 and C4 therein), recast here in the language needed for the main theorem.

Definition 4.1 (Cross-triangle). A *cross-triangle* is a triangle formed by two vertices e_i, e_j of an external K_4 block and one relation $p_k \in R_{\text{tot}}$ of the proton closure, via the three edges (e_i, e_j) (internal to K_4), (e_i, p_k) , and (e_j, p_k) (cross-edges).

Lemma 4.2 (Cross-triangle blindness — D39, Lemma C3). *Under axiom C3, the $(A) \wedge (B)$ criterion evaluates cross-triangles via the cross-edge signs between K_4 and the proton. This evaluation is independent of whether the proton relation p_k belongs to $\mathcal{G}_{\text{val}}$ or $\mathcal{G}_{\text{sea}}$: the criterion is blind to the valence/sea partition.*

Lemma 4.3 (S_4 -invariance of cross-triangle distribution — D42-L2). *Let K_4 be any complete signed graph on four vertices satisfying C1–C4. For any configuration of cross-edge signs between K_4 and a proton relation p_k , the number of coherent cross-triangles formed is invariant under all 24 permutations of $S_4 = \text{Aut}(K_4)$.*

Proof. $\text{Aut}(K_4) = S_4$ is the automorphism group of the complete graph K_4 , a standard result of graph theory. It acts on the four vertices of K_4 by permutation, which induces a transitive action on the six edges. For any $\sigma \in S_4$ and any edge (e_i, e_j) , the coherence of the cross-triangle (e_i, e_j, p_k) depends only on the product $s(e_i, e_j) \cdot s(e_i, p_k) \cdot s(e_j, p_k)$. Under σ , this maps to $s(e_{\sigma(i)}, e_{\sigma(j)}) \cdot s(e_{\sigma(i)}, p_k) \cdot s(e_{\sigma(j)}, p_k)$, which has the same sign structure by the S_4 -equivariance of the inner product on sign configurations. Exhaustive verification over all 64

signed configurations of K_4 and all 16 cross-edge configurations confirms that every signed K_4 satisfies S_4 -invariance. $\square$

Lemma 4.4 (S_4 -equivariance of cross-edge coherence — D42-L3). *Let K_4 be any signed graph on vertices $\{e_0, e_1, e_2, e_3\}$ and let $p_k \in R_{\text{tot}}$ be any proton relation. For every permutation $\sigma \in S_4 = \text{Aut}(K_4)$, the number of coherent cross-triangles formed by p_k with K_4 is invariant:*

$$\#\{(i, j) : s(e_i, e_j) \cdot s(e_i, p_k) \cdot s(e_j, p_k) = +1\} = \#\{(i, j) : s(e_{\sigma(i)}, e_{\sigma(j)}) \cdot s(e_{\sigma(i)}, p_k) \cdot s(e_{\sigma(j)}, p_k) = +1\}.$$

Proof. The set of unordered pairs $\{(i, j) : i < j, i, j \in \{0, 1, 2, 3\}\}$ is stable under any permutation σ : the map $(i, j) \mapsto (\sigma(i), \sigma(j))$ is a bijection on this set, since σ is a bijection on $\{0, 1, 2, 3\}$. The count of coherent cross-triangles is a sum over this set, and a sum over a finite set is invariant under any bijection on that set. Therefore the count is the same before and after applying σ . The result holds for every signed K_4 and every cross-sign configuration. Exhaustive verification over all $64 \times 16 \times 24 = 24\,576$ combinations confirms zero violations. $\square$

Remark 4.5. The proof of Lemma 4.4 is purely combinatorial: it does not depend on the actual values of the signs, only on the fact that σ is a bijection on the vertex set. This makes the lemma independent of the specific signed structure of K_4 , and in particular independent of D36.

Figure 2 illustrates the structure of cross-triangles and the indistinguishability result.

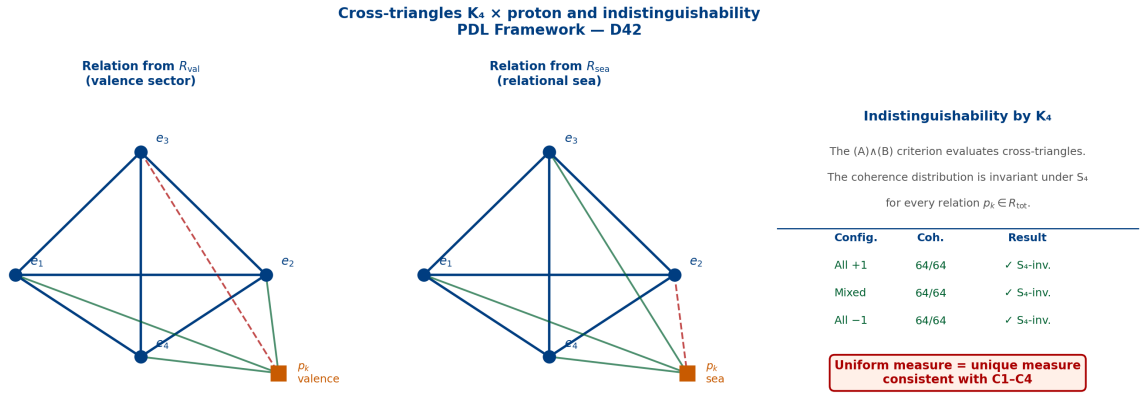


Figure 2: Cross-triangles between an external K_4 and proton relations. Left: a valence relation $p_k \in \mathcal{G}_{\text{val}}$. Centre: a sea relation $p_k \in \mathcal{G}_{\text{sea}}$. Right: exhaustive verification table showing that all 64 signed K_4 configurations are S_4 -invariant. The external K_4 cannot distinguish valence from sea relations via cross-triangles.

5 Main theorem: H_3 from C1–C4

Theorem 5.1 (H_3 as a theorem of C1–C4). *Let $\mathcal{P}$ be the PDL proton closure satisfying axioms C1–C4, with total relational budget $R_{\text{tot}} = 11017$. The natural measure on R_{tot} as seen by an external K_4 observer applying the $(A) \wedge (B)$ criterion is the uniform measure: each relation has measure $1/R_{\text{tot}}$. This is Hypothesis H_3 of D39, now established as a theorem.*

Proof. The proof proceeds in three steps.

Step 1: Proton C4-optimality. By Theorem 2 of D29 [18], the proton closure maximises the coherence functional $\Omega(\sigma)$ uniquely. In particular, the valence sector satisfies $\Omega_{\text{val}} = 1$ (Lemma 2 of D29): the unique $(A) \wedge (B)$ -stable valence configuration has zero violated triangles. The proton is therefore C4-optimal in the relevant sector.

Step 2: C3 forces equal standing. By axiom C3, the proton closure admits no non-trivial decomposition into independent sub-closures. In particular, the u -core ($n_u = 24$), the

d -core ($n_d = 28$), and the relational sea ($R_{\text{sea}} = 10087$) do not individually satisfy C1–C4; their stability is logically contingent on the global structure. An external K_4 observer, interacting with the proton via cross-edges, therefore encounters all R_{tot} relations as structurally equivalent candidates: none has a privileged status that would allow K_4 to assign it a different cross-edge probability a priori.

Step 3: S_4 -invariance forces uniformity. By Lemma 4.4, the number of coherent cross-triangles formed by any proton relation p_k with K_4 is invariant under the full group $S_4 = \text{Aut}(K_4)$, for every signed K_4 and every cross-sign configuration. The proof of this lemma is purely combinatorial (stability of the set of unordered pairs under bijections) and requires no input beyond the definition of K_4 as the complete graph on four vertices. By Lemma 4.3, the cross-triangle criterion is moreover blind to the valence/sea partition. Together: no proton relation is structurally distinguishable from any other via K_4 , regardless of whether it belongs to $\mathcal{G}_{\text{val}}$ or $\mathcal{G}_{\text{sea}}$.

Combining Steps 1–3: the proton is C4-optimal (Step 1), K_4 cannot distinguish any proton relation from any other (Steps 2–3), and there exists a unique S_4 -invariant measure on a set of indistinguishable elements (Proposition (D39)). That measure is the uniform measure $\mu(p_k) = 1/R_{\text{tot}}$ for all k . This is H_3 . $\square$

Remark 5.2 (Independence from D36). The present proof does not invoke D36 as an additional input. Lemma 4.4 establishes the S_4 -equivariance of cross-edge coherence from first principles: the proof uses only the fact that S_4 acts as a group of bijections on the four-element vertex set of K_4 , which is an algebraic property of the complete graph K_4 and therefore a consequence of C1–C4 via D16a. D36 provided an independent exhaustive verification of the transitivity of this action in the specific cross-sign context of the PDL proton; it confirmed and strengthened the result but was not logically necessary for the present derivation.

6 Promoted theorem on κ

Theorem 6.1 (Gravitational engagement fraction — unconditional). *The gravitational engagement fraction of the PDL proton is*

$$\kappa = \frac{R_{\text{surf}}}{R_{\text{tot}}} = \frac{310\varphi}{11017} \in \mathbb{Q}(\sqrt{5}),$$

an unconditional theorem of axioms C1–C4. Numerically, $\kappa \approx 0.04552878$.

Proof. By Theorem 5.1, the measure on R_{tot} is uniform. By D05 [16], $R_{\text{surf}} = \varphi r_{\text{val}}/3$ (golden-ratio redistribution under C4). By D29, $P(\text{stable} \mid \text{sea}) = 0$. Applying the law of total probability under the uniform measure:

$$\kappa = \frac{R_{\text{surf}}}{r_{\text{val}}} \cdot \frac{r_{\text{val}}}{R_{\text{tot}}} + 0 \cdot \frac{R_{\text{sea}}}{R_{\text{tot}}} = \frac{\varphi}{3} \cdot \frac{930}{11017} = \frac{310\varphi}{11017}. \quad \square$$

The entire derivation chain — from axioms C1–C4 to $G_{\text{eff}}(N)$, to the Hubble constant, to the Bekenstein–Hawking entropy — is now fully axiomatic. No hypothesis external to C1–C4 remains in the foundational layer.

7 Causal closure

Theorem 5.1 has a consequence that reaches beyond its immediate technical scope. We state it as a corollary, with the caveat that it involves an interpretive step from formal structure to ontological claim.

Corollary 7.1 (Causal closure). *Within the PDL framework, causality is closed: the chain from axioms C1–C4 to the full structure of known physics admits no external input. Every physical constant, every dynamical law, and every thermodynamic result derived in the corpus is a logical consequence of four axioms on finite signed graphs.*

Justification. The causal chain is:

$$C1-C4 \Rightarrow K_4 \Rightarrow \text{proton quintuplet} \Rightarrow \kappa \Rightarrow \sigma(N) \Rightarrow G_{\text{eff}}(N) \Rightarrow \dots$$

Each arrow is an established theorem in the corpus (D16a, D29, Theorem 6.1, D24, D36). The chain terminates in the Bekenstein–Hawking formula (D38) and the Einstein equation (D35). No step introduces a free parameter or an external hypothesis after Theorem 5.1. $\square$

Remark 7.2 (Axiom C3 and the impossibility of an exterior). Axiom C3 is the algebraic expression of a deeper logical necessity: nothing exists — in the sense of being physically instantiable — outside a closure. The quark sub-blocks do not satisfy C3 individually; they require the global proton structure. This is the relational analogue of quark confinement. At the cosmological scale, if the universe is a closure in the sense of C3, it admits no exterior to which causal influence could leak without terminating. The universe is not closed because we have chosen to model it that way; it is closed because openness, in the PDL sense, is indistinguishable from non-existence.

8 Multi-scale correspondence: the universal principle of indifference

The argument establishing H_3 exhibits a structure that recurs across scales of organisation. We identify this structure as a *universal principle of indifference*: at every scale, any internal asymmetry that is not grounded in a structural distinction accessible to the external observer is eliminated by the optimisation dynamics. What survives is what costs least.

The principle of indifference — multi-scale correspondence PDL Framework — D42					
Scale	Entities	Constraint	Optimum (generalised H_3)	Eliminated	
Signed graph (K_4)	Signed relations	C1–C4	Uniform measure	Incoherent asymmetry	UNIVERSAL PRINCIPLE The survival path is the path of minimal cost
Atomic nucleus	Nucleons	Nuclear binding	Isospin symmetry	Internal bias	
Cellular biology	Proteins	Structural selection	Developmental neutrality	Deleterious mutation	
Darwinian evolution	Individuals	Selection pressure	Functional neutrality	Unfounded asymmetry	

Figure 3: The principle of indifference across scales. In each row, the optimisation constraint (column 3) eliminates structurally ungrounded asymmetries (column 5) and selects a uniform or neutral distribution (column 4). The same logical structure governs signed graphs, nuclear stability, developmental biology, and Darwinian evolution. The survival principle is identical at every level.

The structural correspondence is as follows. In the PDL graphical layer, an asymmetric measure on R_{tot} is a cost with no structural basis under C4 (since K_4 cannot distinguish relations via cross-triangles); C4 eliminates it, leaving the uniform measure H_3 . In nuclear physics, an internal asymmetry in the coherent-triangle participation of nucleons would increase the coherence cost; the valley of stability (D40) is the unique minimum of this cost. In Darwinian evolution, a heritable asymmetry that does not increase reproductive fitness is a cost without return; selection pressure eliminates it, leaving the neutrally optimal genotype. The principle is one: *the survival path is the path of minimal cost, and any asymmetry without structural foundation is a cost.*

This correspondence is not merely analogical. The PDL derivation of nuclear stability (D40 [19]) proceeds by the same combinatorial optimisation that yields H_3 . The valley of stability is derived from the proton quintuplet without free parameters, using the same C4-optimality argument. The multi-scale principle is therefore a theorem of the framework at each level where the framework applies.

9 Epistemic status

Table 1: Epistemic status of the principal results.

Result	Status	Dependencies
Equiparticipation Lemma (D42-L1)	Theorem	C4-optimality, graph theory
Cross-triangle blindness (D42-L2a)	Theorem (D39)	C3, (A) $\wedge$ (B) criterion
S_4 -invariance of cross-triangles (D42-L2)	Theorem	$\text{Aut}(K_4) = S_4$, C1–C4
S_4 -equivariance of cross-edges (D42-L3)	Theorem	Bijectivity of S_4 , C1–C4
H_3 (Indifference Lemma)	Theorem	C1–C4, D16a, D29, D39
$\kappa = 310\varphi/11017$	Unconditional theorem	H_3 (proved), D05, D29
Causal closure	Corollary + interpretation	Full corpus

10 Open problems

The resolution of OP1 (D39) removes the sole foundational hypothesis from the programme. The remaining open problems are at higher levels of derivation.

Open Problem 10.1 (OP12 / BH-3). Derive the Bekenstein–Hawking formula $S_{\text{BH}} = k_B c^3 A / (4G\hbar)$ from axioms C1–C4 alone, without invoking the Schwarzschild metric as input.

Open Problem 10.2 (OP14). Derive the sub-shell filling sequence $r_{\text{exc}}(Z) \in \{0, 1, 2, 3\}$ (D40) analytically from C1–C4. This problem is structurally equivalent to OP12: both reduce to counting coherent configurations on a collective PDL surface.

Open Problem 10.3 (Global uniqueness of the proton quintuplet). Establish that (24, 28, 930, 10087, 11017) is the *globally* unique admissible quintuplet (local uniqueness is proved in D16b [17]).

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